The Data Archive as a Social Network: An Analysis of the Australian Social Science Data Archive

Steven McEachern
Deputy Director
Australian Social Science Data Archive

Overview

- History of the archive
- Understanding social networks
- The data (the metadata??)
- Visualising the network
- Network measures
- What can we learn as archives from social network analysis?

History of the archive

- ASSDA was set up in 1981, housed in the RSSS, ANU to collect and preserve Australian Social Science Data on behalf of the social science research community
  - Now includes nodes at Uni of Melbourne, Uni of Queensland, Uni of WA, University of Technology Sydney, with infrastructure provided by the ANU Supercomputer Facility
- The Archive holds some 2400 data sets, most notable holdings are national election studies; public opinion polls; social attitudes surveys.
- Data holdings are sourced from academic, government and private sectors.
- The Archive also plays a role in the region, helping to re-establish the NZ Data Archive in 2007 and acts as a custodian for countries without data archives.

ASSDA as a social network

- Question: is there value in examining the social network of data archives?
- What could we learn?
  - Theme of the conference – social networks
  - Social network data – often XML, RDF, etc.
  - Parallel with citation networks and co-publication

Understanding social networks

- Social network analysis is focused on uncovering the patterning of people's interaction. It is about the kind of patterning that Roger Brown described when he wrote:
  - "Social structure becomes actually visible in an anthill; the movements and contacts one sees are not random but patterned. We should also be able to see structure in the life of an American community if we had a sufficiently remote vantage point, a point from which persons would appear to be small moving dots. . . . We should see that these dots do not randomly approach one another, that some are usually together, some meet often, some never. . . . If one could get far enough away from it human life would become pure pattern.“
- Freeman, (2008) What is social network analysis? http://www.insna.org/sna/what.html

Contents of a citation social network

- Vertices (points) = authors
- Edges (lines) = co-depositor
  - Can also include number of co-deposits
  - Think of a deposited study as a publication

The data (the metadata?)

- A list of principal investigators from each of ASSDA's ~2400 studies
- Drawn from ASSDA's metadata in Nesstar
  - DDI2.0 Element: A.6.2.1 Authoring Entity (AuthEnty)
  - More accurately – the Nesstar RDF element stdyAuthEntity

Study description

ASSDA Nesstar WebView - Mozilla Firefox

File Edit View History Bookmarks Tools Help

http://assda-nesstar.anu.edu.au/webview/index.jsp?v=2&mode=documentation&submode=abstract&study=http%3A%2F%2Fassda-nesstar.anu.edu.au%3A80%2Fobj%2FfStudy%2F

Most Visited | Latest Headlines | Google Bookmark | Note in Reader | ASSDA - Australian Soci... | ASSDA Basecamp Proje... | Share on Posterous | Google Reader | Clip to Evernote | Import to Mendeley | Post to CiteULike | Readability | TV Guide Advanced Se...

ASSDA - Australian Social Scien... | Sun™ Convergence | Google Reader (1000+) | Australian Social Science Data ... | ASSDA Nesstar WebView

Australian Social Science Data Archive
NESSTAR Service

SEARCH

- Census
- Demography
- Economics
- Education
- Employment, Labour
- Environment, Conservation, Land use
- Health
- Housing
- Industry, Management
- Law, Crime, Courts
- Mass media, Communication, Language
- Politics
- Poll
- Psychology
- Science, Technology
- Social classes, Social order
- Social welfare
- Sociology, Culture
  - Community and urban studies
  - General social survey
  - Leisure, recreation, tourism and sport
  - Membership to organisations
  - Migrants, immigration and multiculturalism
  - Religion, Religious affiliation
  - Social attitudes, values and behaviour
    - Australian Survey of Social Attitudes
      - The Australian Survey of Social Attitudes, 2003
      - The Australian Survey of Social Attitudes, 2005 - Restricted
      - The Australian Survey of Social Attitudes, 2005 (SA)
      - The Australian Survey of Social Attitudes, 2005
      - The Australian Survey of Social Attitudes, 2007
        - Metadata
          - Study Description
            - Bibliographic Citation
            - Study Scope
            - Methodology And Processing
            - Data Access
            - Other Study Description Materials
          - Data Files Description
            - au.edu.anu.assda.ddi.01127
        - Variable Description
        - Bookmarks
    - Australian values study survey, 1983: File nw
    - Australian values study survey, 1983: self-completion questionnaire
    - Men's attitudes towards women, Australia, 1987
    - Community Attitudes Towards Domestic Violence, Australia, 1987
    - The Adelaide Advertiser women's survey, 1988
    - Community Attitudes Towards Smoking in Public Places, 1991
    - Children's Attitudes to Television, 1993: File c
    - Children's Attitudes to Television, 1993: File p
    - R-Classified Programs on Pay TV, 1993
    - Community Attitudes To Road Safety, 1993
    - Community Attitudes to Road Safety, 1995
    - Community Attitudes to Road Safety, 1996

DESCRIPTION | TABULATION | ANALYSIS

Dataset: The Australian Survey of Social Attitudes, 2007

Full Title
The Australian Survey of Social Attitudes, 2007

Identification Number
au.edu.anu.assda.ddi.01127

Authoring Entity
- Phillips, Timothy Affiliation: The Australian National University
- Mitchell, Deborah Affiliation: The Australian National University
- Tranter, Bruce Affiliation: University of Tasmania
- Clark, Juliet Affiliation: Deakin University
- Reed, Ken Affiliation: Deakin University

Other identifications and acknowledgments

Producer
- Australian Demographic and Social Research Institute Affiliation: The Australian National University Abbreviation: ADSRI Role: Financial and administrative responsibility

Copyright
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Date of Production
26 June, 2008

Place of Production
Australian Social Science Data Archive

Data Distributor
- Australian Social Science Data Archive Abbreviation: ASSDA
  Affiliation: The Australian National University

Contact Person
- Rachelle Graham
  Australian Social Science Data Archive
  18 Balmain Crescent
  The Australian National University
  Acton ACT 0200
  Affiliation: ASSDA

Depositor
- Ann Evans
  Affiliation: The Australian National University

Date of Deposit
2008-06-26

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POWERED BY nesstar

Done

What does the data look like?

Bruce Headey Alexander J Wearing

Homel, R. Lecturer, S.

Hamilton, I. Peterson, T.

Jaensch, D. Loveday, P.

NSW Bureau of Crime Statistics and Research

Department of Community Services and Health

Australian Bureau of Statistics

Saulwick Research

Scott, W. A. Scott, R.

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Data transformation

- Need a file with separate authors, and their links to other authors

- Data is actually stored as text (CDATA?)
- Separation out of separate authors
- Reordering into consistent author format
- Generation of author links (a variation on moving from wide to long format, but with multiple iterations across the multiple author relationships in a study)

Final data format

*Vertices 644

1 "Ada, A.”

2 "Adams, Kathryn”

3 "Aimer, Peter“

4 "Aitkin, Donald“

5 “Alexander, I.”

6 “Alexander, M.”

…

Final data format

*Edges

2 21 8

2 528 8

3 279 1

3 280 1

4 42 1

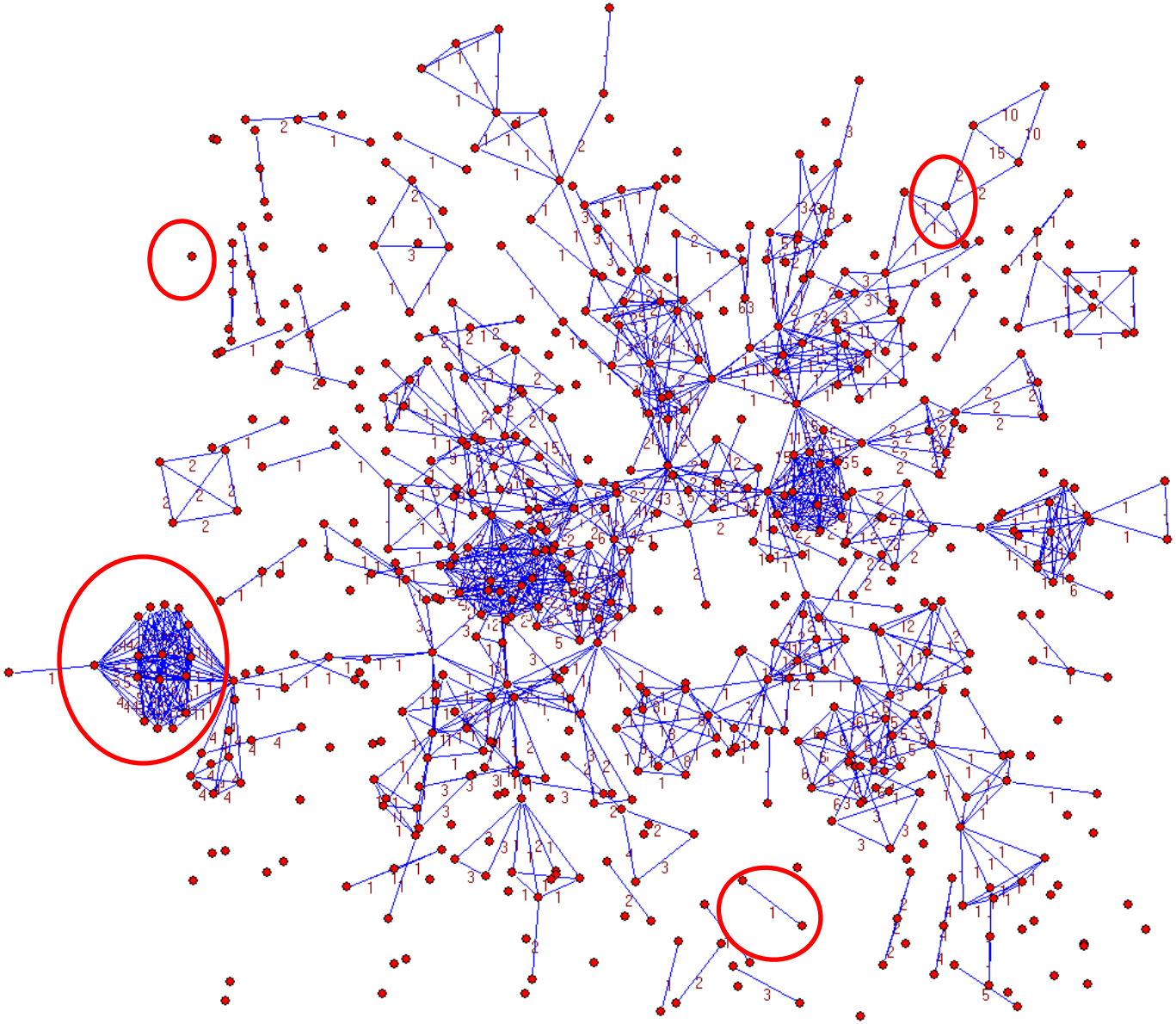
4 104 1

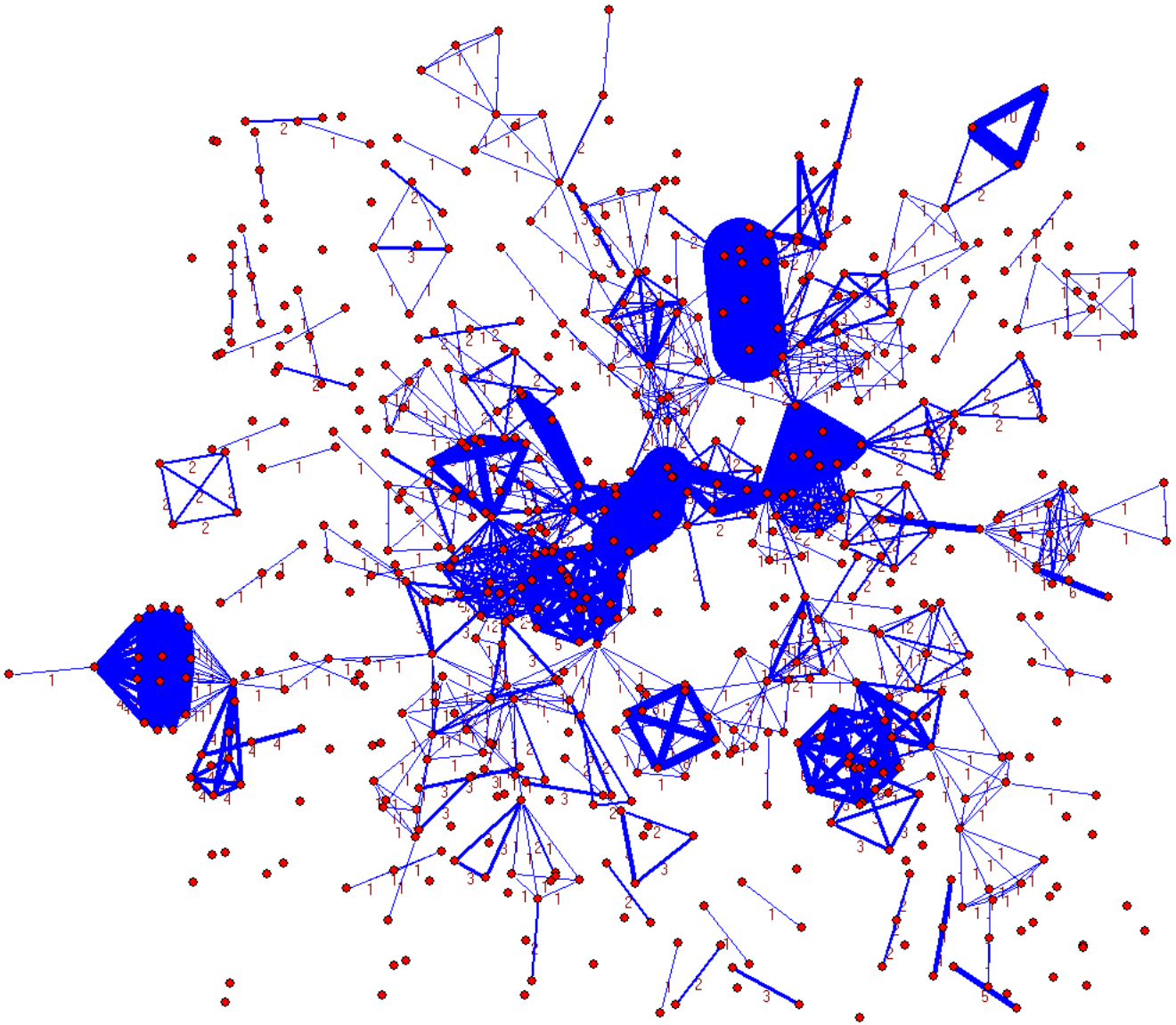
4 237 1

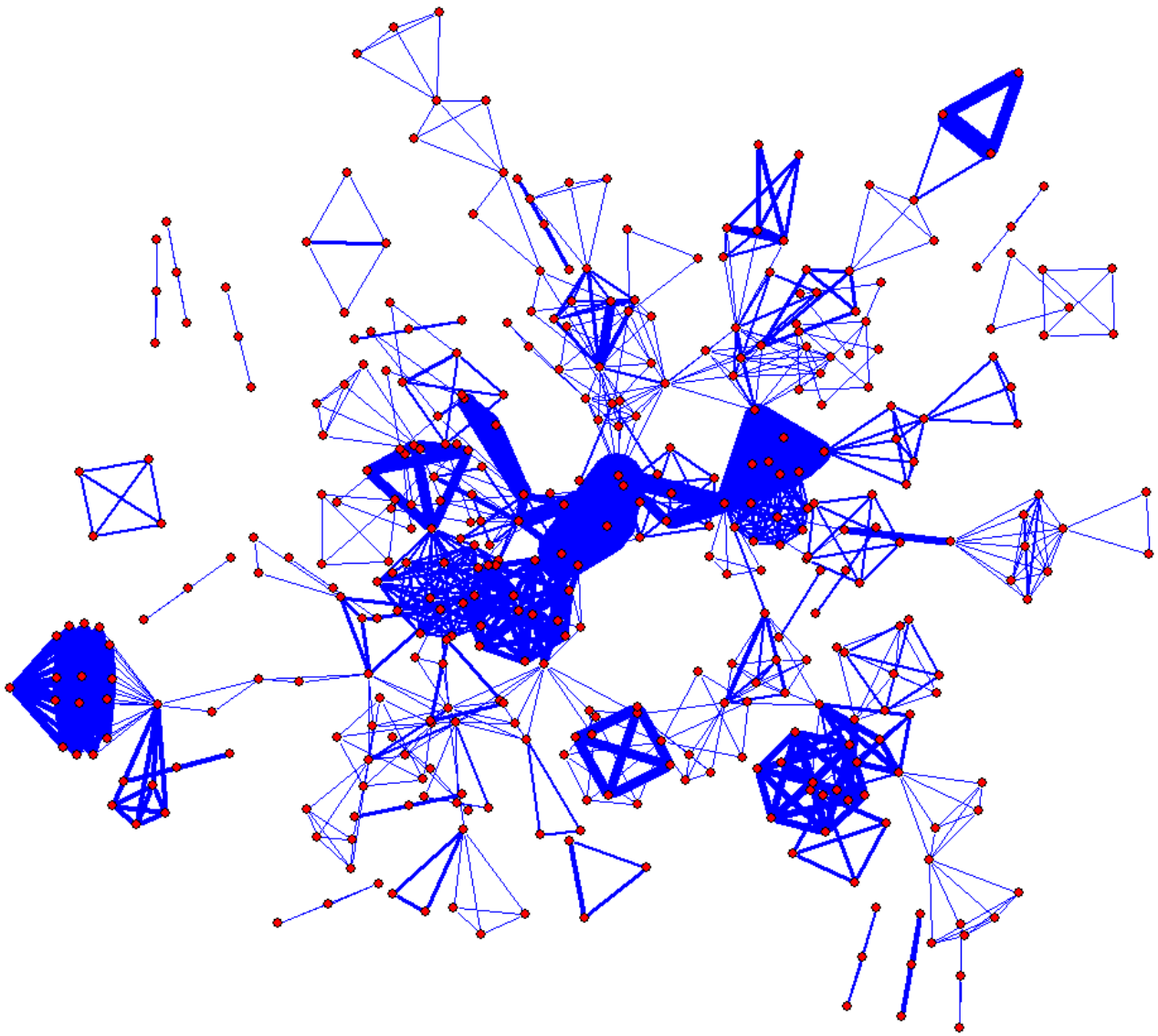
1st author, 2nd author, number of common studies

Visualising “ASSDAnet”

- Visualisation software: Pajek
  - Free software for visualisation of large social networks
- Statistical software: R
  - Pajek has an export plugin for porting directly to R







Network measures

Node measures

- Degree: number of edges for the vertex
- Betweenness:
  - Betweenness measures the extent to which a given vertex lies on non-redundant geodesics between third parties
- Closeness: “average” (geodesic) distance between a vertex and all other vertices
  - not useful in situations such as this – have some isolated nodes i.e. indiv. depositors

Degree

Lee, Christina	48	Korten, Ailsa	32
McAllister, Ian	44	Macintyre, Clement	32
Smith, Anthony	42	Mackinnon, A.	32
Bean, Clive	40	Olds , Timothy	32
Bowen, Jane	32	Syrette, Julie	32
Burnett, Jill	32	Luczsz, Mary	30
Cobiac, Lynne	32	Vowles, Jack	30
Dollman, James	32	Western, John	30
Jones, Roger	32	Brown, Wendy	28
Jorm, Anthony	32	Byles, Julie	28

Betweenness

Bean, Clive	Western, John
Lee, Christina	McDonald, Peter
McAllister, Ian	Jones, F.
Makkai, Toni	Korten, Ailsa
Gibson, D.	Goot, E.
Western, Mark	Headey, Bruce
Kendig, H.	Gibson, Rachel
Smith, Anthony	Duncan-Jones, P.
Mackinnon, A.	Henderson, A.
Vowles, Jack	Wearing, Alexander

Network measures
(Butts, 2008)

Graph measures

- Density: 0.0052 (low density)
  - “the fraction of potentially observable edges which are present in the graph”
- Reciprocity: 1.0002 (low reciprocity)
  - “fraction of dyads which are symmetric (i.e., mutual or null)”
- Transitivity: 0.6885 (moderate)
  - Presence of triadic relationships (tendency for A and C to be linked where AB and BC links also occur) – note codepositor clusters

Lessons from SNA

- Simple visualisation shows clustering of co-depositors in the archive
  - Most commonly, multiple deposits of waves of a study by multiple Pis
- Can also see high number of “isolated” depositors
  - Usually institutions – who don’t list Pis
- Measures of centrality can assist with showing linking depositors: those depositing with multiple, independent colleagues
- Might enable targetting of social networks of regular depositors
  - Would be particularly assisted when accompanied by data citation programs (eg. DataCite, King and Altman)

Where to next?

- Two-mode network: depositors by institution
- Time-lapse network: depositors by institution by time
- Cross-national networks??
- Similarity of deposit and publication networks

Website/ Contact

Australian Social Science Data Archive
18 Balmain Crescent
The Australian National University
ACTON ACT 0200

Email: assda@anu.edu.au,
Website: www.assda.edu.au
Phone: +61 2 6125 2200
Fax: +61 2 6125 0627